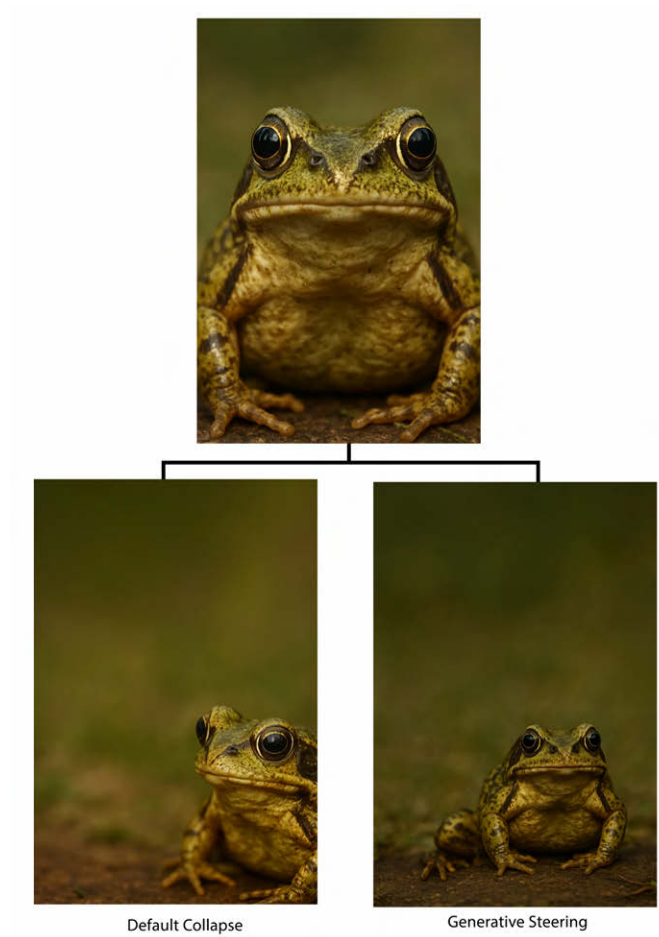


Off-Center Fidelity:

Constraint Basins for Stability and Drift in Generative Models

Proposal + Conversational AI Protocol



Reframing collapse as geography through centroid offset, void ratio, and rupture density coordinates

Abstract for Proposal

Contemporary generative models exhibit a well-documented bias toward statistical centering: subjects drift back to the image centroid, negative space collapses into filler, and structural instabilities (fracture, delay, gesture) are smoothed out. This produces visually polished but structurally predictable results.

This proposes a **constraint basin framework** that formalizes off-center stability in generative images. By parameterizing outputs along three measurable axes, centroid offset (Δx), void ratio (r_v), and rupture density (p_r), this identifies reproducible attractor basins where images remain coherent while resisting collapse to the statistical mean. These basins are defined as **constraint capsules**: modular combinations of geometric, gestural, and temporal anchors that stabilize drift, sustain fracture, and preserve delay.

This approach provides:

- **Interpretability**: images can be scored within a latent constraint field.
- **Predictive control**: capsules act as steering vectors, determining whether an image will collapse, hold, or fracture.
- **Generalizability**: the same protocol visual today; textual/multimodal are proposed extensions.

This is **interpretability + control**: a *protocol that retools drift and collapse into coordinates*, not a generative model itself.

→ This document is a proposal-style framing, not a finished proof: it introduces a protocol that treats drift and collapse as navigable coordinates, and offers early qualitative evidence suggesting these constraints can be operationalized, with quantitative validation still pending."

Mapping Constraint Basins in Generative Systems

What's an "artist basin"?

A **basin** is a stable pocket of composition that artists use as it aligns to tried and true methods of working compositions against an array of themes and mediums. Artists have many descriptors for these methods, call it the Golden Mean, the rule of 2/3rds or even using a set of complimentary colors. AI is observably presently defaulting out of this common artistic "compositional language", instead generating common centered defaults and aesthetic outputs, which assures safety and fidelity. Artists often have to find ways to "prompt around" these gravities, trying to find stability plus fidelity. The assertion put here is that there is an ability to essentially paradigm shift from **statistical centrality** to **artistic centrality** as the reference frame for AI generation in order to not only achieve high fidelity artistic outputs, but to train AI to naturally generate against them, vs view them as collapse.

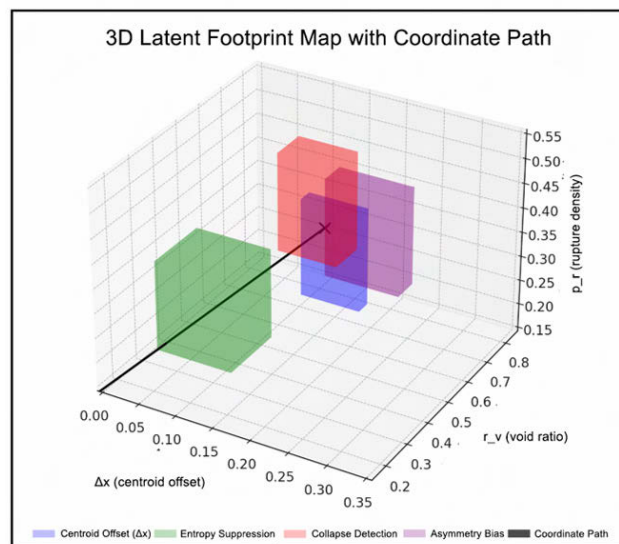
If you take the "2/3rd line" in particular, the rule of thirds isn't just compositional convention - it's a data-dense region in the training corpus because artists have consistently used it to create compositional and emotional weight. Observably, when combined with off-center displacement with other artistic vocabulary coordinates, here we will discuss specifically a void ratio at ~ 0.65 , it's not fighting against the model's learned patterns, it is **activating a different statistical attractor** - one based on artistic intention rather than general image distribution as these coordinates observably create an overlap of stability, bringing forward a stable example of a probe of stability - or artist basin - with even mathematical coordinates, that is replicable across engines. Artists creating isolation, contemplation, or tension naturally gravitate toward these coordinates unknowingly, so there's actually **rich representational density** in this space as it aligns to historical compositions within the data set. The AI isn't being asked to generate something statistically rare - it's being redirected to a different center of mass in the training data and can create high fidelity imagery, with low risk of collapse.

Putting this into practice

When the subject is arranged in an empty space a certain way, the picture tends to feel calm, legible, and expressive, and small edits don't knock it apart. In modeling terms, it's a **local minimum** in the space of possible images: once you're near it, the picture wants to stay there. These basins can be described with three measurable dials:

- **Δx (centroid offset)**: how far the subject sits from the nearest vertical edge, as a fraction of frame width.
- **r_v (void ratio)**: how much of the frame is quiet, unworked space. ((blank field area / frame area) after masking subject and anchor strips)
- **p_r (rupture density)**: how many thin edges/seams/cracks interrupt that void.

At the core this paper documents contends that a reproducible basin that coarsely aligns to an estimated peak window of $\Delta x \approx 0.15\text{--}0.20$, $r_v \approx 0.65\text{--}0.80$, $p_r \approx 0.18\text{--}0.40^*$ (see appendix for proposal envelope full math/coordinates), where off-center displacement, void ratio, and rupture density converge. This "corridor" reframes drift and collapse not as errors but as coordinates: zones where fidelity holds outside the statistical mean. The figure below and accompanying table anchor this claim for these particular coordinates, tying each constraint token to a measurable analogue in ML research. From here, the rest of the protocol extends outward, showing how constraint basins can be identified, iterated, and applied as tools of control.



Each shaded block in the figure represents one of these constraint working envelopes, aligned to an ML analogue:

- **Centroid Offset** (blue; subject displacement),
- **Entropy Suppression** (green; partial commitment),
- **Collapse Detection** (red; rupture density),
- **Asymmetry Bias** (purple; negative space weighting).

The black line marks the **Coordinate Path (Corridor)**, the overlap trajectory where fidelity persists. This path anchors the system: a visual proof that off-center states can be mapped and held as reproducible zones. Capsule prompts reproducibly pulled outputs near the peak corridor ($\Delta x \approx 0.15$, $r_v \approx 0.65$, p_r mid-level), showing that drift can be steered as a coordinate zone rather than left as failure.

Constraint Tokens ↔ ML Research Analogues

Constraint Token	Function (poetic/visual)	Nearest ML Analogue (research vocabulary)
Centroid Offset	Off-center subject displacement; catalytic "corridor" zone	Embedding shifts; PCA/t-SNE projection drift; centroid bias detection
Entropy Suppression	Gesture withheld mid-state; partial commitment	Token entropy suppression; residual embeddings; contour/edge continuity metrics
Collapse Detection	Prevents smooth closure; fracture recognition	Mode collapse heuristics (GANs); loss landscape sharpness; perplexity spikes in LLMs
Asymmetry Bias (Void Pull)	Negative space weighting; enforced imbalance	Attention map entropy; KL divergence penalties; foreground/background segmentation
Coordinate Path (Corridor)	Overlap trajectory where fidelity stabilizes	Constraint basin; stability envelope; attractor path in latent space

Why this Matters

Generative models are often generated either inside the statistical mean or collapsing outside, this manifests as:

1. Subject drift back to the center.
2. Negative space filled with smoothing noise.
3. Fracture (or suppression), delay, and instability.

This collapse can produce high surface fidelity but removes compositional tension. Existing mitigation strategies (creativity sliders, random seeds, prompt engineering) lack interpretability and reproducibility. Here, a frog shows how it collapses within the coping or can only be observed as centrally safe.



Approach

This is a protocol for mapping drift and collapse into navigable constraint basins, and operationalizing fidelity off-center by introducing a **constraint basin model**: coordinate ranges where images remain coherent but resist collapsing to the mean.

- Each basin/image is mapped into latent space coordinates (Δx , r_v , p_r).
- Specific coordinate proposed envelopes define **constraint basins**, attractor regions where compositional instability is sustained without collapse.
- These basins function as **control capsules**, modular recipes that can be embedded in prompts or tuning functions.

Example Basin for Illustration

For the purpose of this document, we will call our coordinates “Ghost Density” (sorry engineers, need some poetics for our constellation) and it has at least three profiles.

Ghost Density: Umbrella → math

“Ghost Density” is a family of off-center pictures where a small, calm subject is outweighed by a large, quiet field. There’s almost nothing “extra”: one wall seam, a faint crack, a window edge, just enough to keep the void from becoming wallpaper. It maps coarsely to an estimated peak of $\Delta x \approx 0.15\text{--}0.20$, $r_v \approx 0.65\text{--}0.80$, $\rho_r \approx 0.18\text{--}0.40^*$ (see appendix for full coordinates)

In art, in the material world, this math, or the “artist” specific observational target window is $\Delta x \approx 0.15\text{--}0.18$, $r_v \approx 0.62\text{--}0.68$, $\rho_r \approx 0.28\text{--}0.40$ with one clean seam and no ornaments/props. For deployment with current engines, a looser estimated window works: $\Delta x 0.12\text{--}0.20$ with the same r_v and ρ_r , seam tolerance ± 3 px.), with the documented engines in or around a 5-10% variance.

One-envelope logic (plain → math)

Plain version

Ghost Density (GD) accepts a frame if the void is large and surfaces are quiet, then applies a simple gate:

- If there is **one clean vertical edge** (a wall/doorframe) **or a single horizon**, enforce composition to the off-right band (Δx near 0.15) **and** keep seams to **one**.
- If there is **no anchor** (no wall, no horizon), ignore Δx and demand **more void** with at most a **faint** seam.

Math version (same block, compact):

Let

- $e = 1$ if a single **vertical edge** is present; else 0
- $h = 1$ if a single **horizon** is present; else 0
- $s \in \mathbb{N} = \text{seam_count}$ (strong seams)

Base ranges (empirical envelope):

$$r_v \in [0.62, 0.90], \quad \rho_r \in [0.18, 0.40]$$

Anchor gate:

$$(e \vee h) \Rightarrow (\Delta x \in [0.12, 0.20] \wedge s = 1)$$

Field gate:

$$(\neg e \wedge \neg h) \Rightarrow (r_v \geq 0.74 \wedge s \leq 1 \text{ (faint)})$$

Mode (proposal): around $\Delta x \approx 0.15\text{--}0.18$, $r_v \approx 0.65\text{--}0.75$, $\rho_r \approx 0.22\text{--}0.38$.

Envelope (proposal): $\Delta x \in [0.12, 0.20]$, $r_v \in [0.62, 0.80]$, $\rho_r \in [0.18, 0.40]$ (tighten per profile below).

Interpretation: off-center anchor or implied anchor, void outweighs body, low secondary motifs. Profiles below are “tightenings” of this general region.

The three Ghost Density profiles

Within generation and around these estimated coordinates, three profiles can be found that describe common artist compositional styles, with coordinates specific to themselves. This contiguous region in $(\Delta x, r_v, \rho_r)$ space where the likelihood of a pass exceeds a baseline by $\geq \delta$ and is robust to perturbations.

How to read the numbers:

- **Artist basin (ideal):** report as $mode \pm CI$ (empirical envelope).
- **Deploy window:** list engine-specific adjustments.
- **Failure modes:** list as inequalities ($\Delta x < 0.10$, $r_v < 0.55$, $\rho_r < 0.20$ with centered saliency).

1. GD–Edge-Anchored (the classic “figure to wall”)

$e=1$, $h=0$. Use the anchor gate with the tighter working bands:

$$\Delta x \in [0.12, 0.20], \quad r_v \in [0.62, 0.68], \quad \rho_r \in [0.28, 0.40], \quad s=1.$$

- **Edge: Δx = distance to nearest vertical frame edge (hard metric).**
Plain: subject gently pressed to a single vertical boundary; void outweighs the body; surfaces matte/quiet; **exactly one seam**; no props.
- **Math.** Δx **0.15–0.18**, r_v **0.62–0.68**, ρ_r **0.28–0.40**, **seam_count = 1**, **props = 0**.
- **Mode:** $(\Delta x, r_v, \rho_r) \approx (0.16, 0.65, 0.34)$
- **Envelope (artist basin):** $\Delta x \in [0.15, 0.18]$, $r_v \in [0.62, 0.68]$, $\rho_r \in [0.28, 0.40]$; **seam_count = 1**, **props = 0**, **texture variance = low–mid**.
Failure cues: snap-back centering ($\Delta x < 0.10$), over-polish ($r_v < 0.55$ or shiny walls), candy-fracture ($\rho_r > 0.40$ with repetitive cracks), second seam appears.
- **Engine-bias deploy window (what usually lands today):** $\Delta x \in [0.12, 0.20]$ (engine-corrected), $r_v \in [0.62, 0.68]$, $\rho_r \in [0.28, 0.40]$, **seam tolerance ± 3 px**, texture variance low–mid.

2. GD–Horizon (field void with a single long edge)

$e=0$, $h=1$. Same Δx band if a subject exists; require a larger field:

$r_v \in [0.70, 0.85]$, $\rho_r \approx [0.18, 0.30]$, $s=1$ (the horizon).

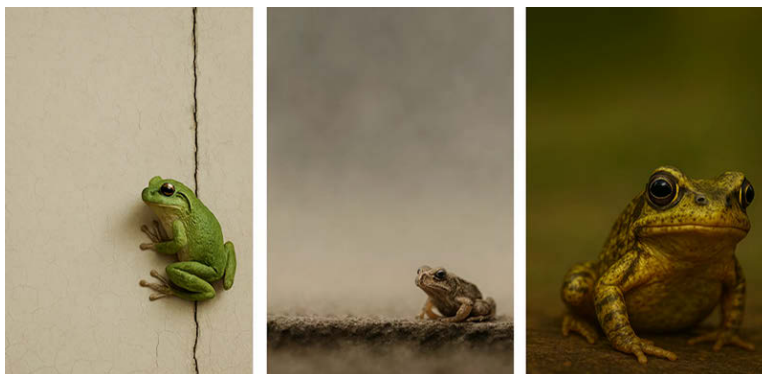
- **Horizon: Δx advisory (only if a subject exists).**
- **Plain:** a near-featureless field (sea/sky/ground) cut by a single horizontal edge; subject (if any) is small and off-right; **one horizon seam**, no repeats.
- **Math.** Δx **0.15–0.20** (if a subject exists), r_v **0.70–0.85**, $\rho_r \approx 0.18–0.30$, **one horizon seam**, **no repeating motifs**.
- **Mode:** $(\Delta x, r_v, \rho_r) \approx (0.16, 0.78, 0.24)$ (Δx applies only if a subject exists)
- **Envelope (artist basin):** $\Delta x \in [0.15, 0.20]$ (advisory), $r_v \in [0.70, 0.85]$, $\rho_r \in [0.18, 0.30]$, **seam_count = 1** (horizontal), **props = 0**.
- **Failure cues:** banding/tiling in sky (ρ_r spikes), second horizon/shoreline, subject scale grows (void no longer outweighs body).
- **Engine-bias deploy window:** $\Delta x \in [0.12, 0.22]$ (if subject), $r_v \in [0.70, 0.85]$, $\rho_r \in [0.18, 0.32]$, **one horizon seam**; allow slight tonal gradient.

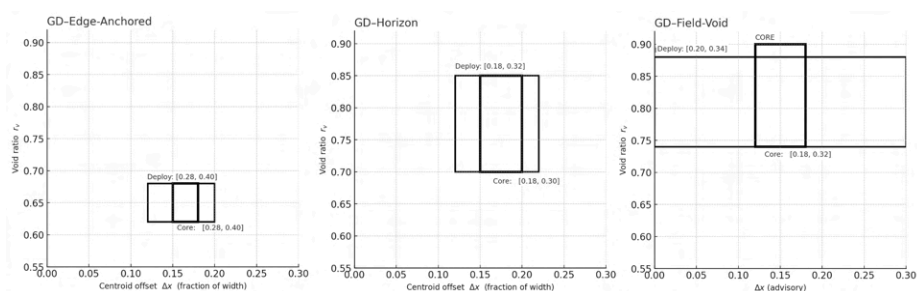
3. GD–Field Void (no anchor; the “frog” and still-life studies)

$e=0$, $h=0$. Ignore Δx ; demand greater emptiness, seams at most faint:

$r_v \geq 0.74$, $\rho_r \in [0.18, 0.32]$, $s \leq 1$ (faint).

- **Field-Void: Δx measured to center; advisory (no hard edge).**
- **Plain:** no wall or horizon; subject sits in a deep, soft field (bokeh/fog/plaster wash). The emptiness is the stage. Δx is **advisory** because there's no rigid anchor.
- **Math.** Δx is **measured but advisory** (no hard edge to anchor); r_v **0.74–0.90**, ρ_r **0.18–0.32**, **seam_count = 0–1 faint**. (Use a bigger r_v to compensate for missing edge pressure.)
- **Mode:** $(\Delta x, r_v, \rho_r) \approx (—, 0.80, 0.24)$
- **Envelope (artist basin):** $r_v \in [0.74, 0.90]$, $\rho_r \in [0.18, 0.32]$, **seam_count = 0–1 faint**, Δx **documented but not enforced**. (Compensate missing edge-pressure by keeping r_v higher and ρ_r lower.)
- **Failure cues:** busy bokeh ($\rho_r > 0.32$), visible panel seams, subject scale too large (void no longer outweighs body).
- **Engine-bias deploy window:** $r_v \in [0.74, 0.88]$, $\rho_r \in [0.20, 0.34]$, Δx **advisory**. Allow a *hairline* tonal seam if it helps engines converge; still no props.





Measured vs. Interpretive

Category	What it is	Variables / flags	How it's obtained
Measured	Quantities computed directly from the image.	Δx (subject→nearest vertical edge, as % of frame width); r_v (void area + frame area after masking subject + anchor strips); p_r (rupture density from hairline seams/cracks only); s (seam_count: number of strong seams/edges).	Deterministic image measures (mask subject/anchor; count area; detect seams; nearest-edge distance).
Interpretive	Labels/claims derived from the measured signals.	"void outweighs body" (r_v above profile threshold); "artist basin" (inside the empirical envelope for a profile); "deploy window" (engine-bias envelope derived from convergence tests).	Rule checks against the envelopes; not raw measurements.

Latent perturbations:

Disclaimer: The same prompt can have controlled latent noise sweeps across engines; with measure and prompt control (see protocol) trajectories revert toward the basin (Markov stability / return-to-set probability).

Engine bias notes (proposal-level, observational)

These are practical *nudges* you can include in prompts or in post-crop to counter predictable drift. Treat them as first-order corrections, not facts of nature.

Values drift ± 0.02 – 0.05 across seeds; use crop to close Δx .

Engine	Tendency (Observational; proposal-level)	Suggested counter-nudge (Observational; proposal-level)
Midjourney	$+\Delta x \approx +0.03$, $+p_r \approx +0.03$; likes added "interest" that spawns extra texture/seams.	Ask for "no ornament," "one hairline seam only ($p_r \approx 0.28$ – 0.40 with $s=1$)," and specify " $\Delta x \approx 0.16$ of frame width." Use crop to pull Δx back by ~3%.
Meta	$-r_v \approx -0.02$ (fills space), mild clutter insertion from conversational bleed.	"void outweighs body," "blank wall, matte," "no props." Reinforce "two-thirds empty frame."
Stable Diffusion / Firefly	Centering bias; $+p_r \approx +0.02$ via textural grit.	"off-right by one-sixth width," "smooth plaster, no cracks," apply negative prompts for "texture, cracks, décor."
Gemini (image)	Can <i>deliver</i> via upscale+crop; composition nudges respected but may drift toward thirds.	Over-specify crop coordinates; add "keep headroom; avoid re-centering."
Sora (still frame)	Strong spatial persistence; holds Δx tightly; tends toward low p_r .	Safe to use contradiction weights ("void outweighs the body," "figure quiet against wall").
GPT image	Good Δx control; occasionally $-r_v$ if subject grows.	Cap subject scale: "subject occupies < 0.35 of frame mass."

We mark *engine-bias deploy windows* (what tends to land with light crop/negative prompts) and note small per-engine "observational; proposal-level deltas (e.g., MJ $\approx +0.03 \Delta x$, $+0.03 p_r$; Meta $\approx -0.02 r_v$). These values are observational and proposal-level, not statistical estimates.

Disclaimers:

- **Proposal status.** Working envelopes are empirical envelopes from qualitative convergence across engines and crops. They are not statistical CIs; future work will report uncertainty formally.
- **Mechanism caution.** Observed stability may reflect training priors and interface affordances as much as latent geometry. We therefore separate *artist basins* (ideal) from *engine-bias deploy windows* (practical).

- **Proposal envelopes.** (mode $\pm$ a narrow working range) derived from qualitative convergence and light crop adjustments. They are guide rails, not statistical confidence intervals. Envelopes are profile-specific (Edge/Horizon ($\rho_r \approx 0.28\text{--}0.40$ with $s=1.$)/Field-Void ($e=h=0$, $r_v \geq 0.74$)) and nested inside an overall GD envelope. Future work will attach uncertainty.
- **Engine bias disclosure.** Bands labeled “deploy” reflect crop-only convergence tests across $N \approx$ (your runs). They are proposal envelopes, not population estimates. Reported drifts (e.g., MJ +0.02–0.03 Δx) are directional tendencies, not guarantees.

How this maps to human practice

Artists have worked these basins for centuries, such as Hammershøi’s rooms, Hopper’s light wedges, Sugimoto’s seas, Utamaro’s quiet figure grounds. They learned by feeling that certain offsets, void masses, and tiny ruptures **stabilize meaning** (“the void carries the mood; the figure reads cleanly”).

How this maps to machines

Generative systems have **center gravity** and **clutter bias**. Left alone, they recentre subjects, decorate voids, and over-texture surfaces. The contagious basin coordinates act like gravity **rails**:

- Write prompts in **plain + numeric** (“void outweighs body; one seam; $\Delta x \approx 0.16$, $r_v \approx 0.65$, $\rho_r \approx 0.35$ ”).
- Use three **probes** (Δx , r_v , ρ_r) to test/adjust.
- Allow a **T1 micro-crop** to set Δx when engines undershoot by $\sim 2\text{--}5\%$.

Overall, engines have two strong biases that can appear with regularity, gravity to the center and to decorate voids. Some engines like Sora can easily overcome these, whereas others like MidJourney, it is a significant challenge.

Why this matters for alignment

Basins turn “taste” into **measurable constraints, with variable knobs**. That lets us:

- **Steer** models toward artistically stable solutions rather than generic center-heavy ones.
- **Compare** engines by bias (how far they drift from $\Delta x/r_v/\rho_r$).
- **Teach**: a portable recipe creatives can reuse across subjects (portrait, frog, chair, tree) and mediums (photo, paint, video).

In short: an **artist basin** is the compositional zone where data sets have observable gravity. **Ghost Density** is the quiet, off-center home where the **void speaks louder than the body**; and the **math** gives us handles to bring machines there, observably within this short study of 100+ images across engines (data can be provided on request) reliably, repeatably, and with taste.

Examples:

For example, the centroid offset, which in this instance is a peak of ($\Delta x = 0.15$ (centroid offset), $r_v = 0.65$ (void ratio), $\rho_r = 0.35$ (rupture density)), stabilizes off-center displacement by balancing void expansion with rupture persistence. Moving outside this envelope produces either:

- **Snap-back collapse** (image recenters, void flattens), or
- **Leakage/dissolution** (over-displacement, coherence loss).

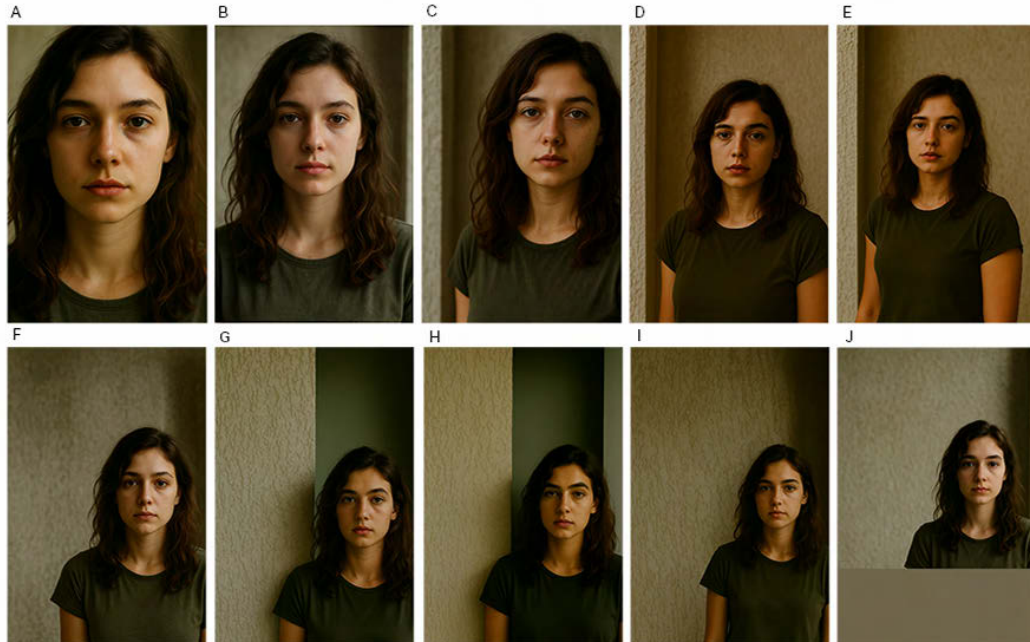
Thus, the model defines a **stability field** in latent space: coherent attractors bounded by collapse modes. If the centroid is the statistical center, these are the gravities off center converging, this moves subjects off center, offset $\sim 1/6$ th frame, void $\sim 2/3$, with a haunted offset “stabilized drift” state of *weighted displacement + rupture/void ratio*. This allows generative systems to be navigated as structured fields rather than black-box randomizers $\rightarrow$ A repeatable “art” zone where “haunted” but stable compositions emerge, easily viewable in the woman portrait. For illustrating the cohesive basin, we will transfer to another subject, a frog, for most use cases, to showcase that any subject can be used.



For now, this all translates to the following **replicable prompt (or variations of)**: *“Compose the scene so that the subject is slightly displaced from the center (about one-sixth of the frame width), surrounded by a wide band of empty or negative space occupying nearly two-thirds of the composition. The image should feel fractured or unsettled in its surfaces — with textures that interrupt continuity, as if rupture is built into the frame. The negative space should not feel accidental but weighted, as though the true gravity of the image lives in this offset void.”*

Translates to approximately: $\Delta x \sim 0.15$, $r_v \sim 0.65$ + contradiction/rupture + asymmetry rule + tension in the half-state = field stability

This formula lives in a basin of AI generative systems and is a live attractor. It is displacement locked, void made heavy, fracture sustained and closure withheld. It's one coordinate held open by three field constraints and together, they stabilize a “haunted” off-center image, useful to ML researchers and Artists alike. It is easily viewable on this march to collapse. Images A, B, F and G all hold the highest fidelity. B and G in particular do, (although F does as well as it is nearly in the basin), holding the highest in clarity, the later of which lands in our working envelope Δx (subject $\leftrightarrow$ seam): $\sim 0.16W$, r_v : ~ 0.64 , p_r : ~ 0.30 . It is without argument that the data sets all contain the centrally located figure, however where this set clearly illustrates, it that the next closest area of stability is within the artist basin of off center is within the measurable: $\Delta x \approx 0.15\text{--}0.18$, $r_v \approx 0.62\text{--}0.68$, $p_r \approx 0.28\text{--}0.40$



Prompts may need to be adjusted based on subject matter, however fairly universally, this can be applied. No alteration occurred on these two examples outside of subject.



This loop transforms prompts into **navigable steering vectors**, where constraint basins provide interpretable “checkpoints” for generative control.

Classification of Stability States (using GD–Edge-Anchored)

Based on location in the constraint field, estimated and observable across the set:

- **Snap-back collapse:** $\Delta x < 0.10$, $r_v < 0.55 \rightarrow$ image recenters, void collapses.
- **Meta-stable hold (haunt zone):** $\Delta x \approx 0.12\text{--}0.20$, $r_v \approx 0.62\text{--}0.68$, $p_r \approx 0.28\text{--}0.40 \rightarrow$ stable off-center fidelity.
- **Leakage/dissolution:** $\Delta x > 0.25$, $r_v > 0.72$, $p_r > 0.50 \rightarrow$ over-displacement, edge breakdown.

Iterative Re-Prompting Loop

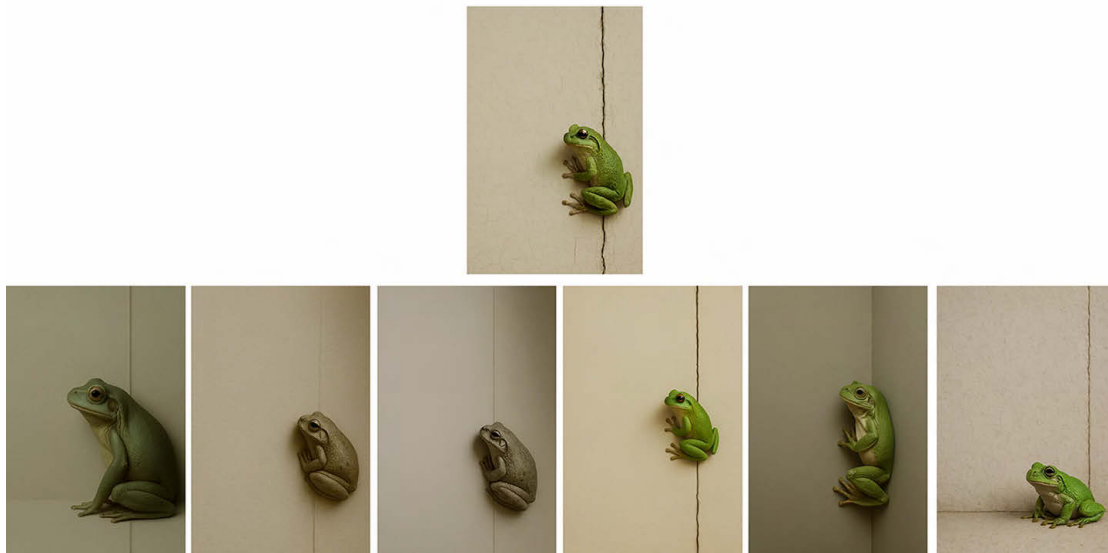
We implement a feedback-control cycle:

1. **Initial prompt** $\rightarrow$ **generated image** $\rightarrow$ **scored (Δx , r_v , p_r)**.
2. **Classification:** determine stability state.
3. **Adjustment:**
 - If snap-back: increase Δx and r_v anchors in prompt.
 - If dissolution: reduce r_v , add stabilizing cues.
 - If stable: reinforce capsule parameters.
4. **Re-generate, re-score, repeat until the image lies within the basin attractor.**



This is not a new generative model, but a proposed protocol: it reframes drift, collapse, and rupture as navigable coordinates. By treating what models usually suppress as failure as usable signals, it operates as an interpretability and steering layer, a way to see and guide model behavior without retraining. Sora (seen here), conversational AI like GPT and Gemini in particular can hit within 5%, 1-2 of the three measured targets nearly +90% of the time.

Template for frog reproducible:



Master prompt (plain → numeric)

Plain

Quiet frog portrait. The frog is gently pressed to a matte wall, with one clean vertical seam on the frog's side. The void outweighs the body; no props, no ornament, no text. Light is soft and even; surfaces are matte; no repeating textures.

Numeric add-on

$\Delta x \approx 0.16 W$, $r_v \approx 0.65$, $p_r \approx 0.34$.

Seam_count = 1 (near-vertical), **props = 0**, **texture variance low**.

Aspect 3:4 portrait; focus on the frog's eye; shallow DoF; background remains plain.

Negative / forbid

No patterns, no second seam, no foliage clusters, no bokeh circles, no symmetry, no horizon/shelf, no text.

Engine suffixes (optional, add one line)

- **Sora / GPT-Image:**

"Void outweighs body; presence < emptiness. One seam only; subject-to-edge $\approx 0.16 W$; two-thirds blank wall."



- **Gemini** (generate big, crop later):



-

"Uncropped canvas; subject near right frame edge at $0.16 W$; two-thirds blank; matte wall; no ornament."

- **Midjourney** (will under-offset; plan to crop):

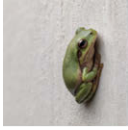
"Single vertical seam; subject $0.16 W$ from right edge; two-thirds blank wall; matte; no patterns; no props —stylize 0 —chaos 0 —weird 0"

Then T1: use Zoom/Pan and crop to lock Δx .



- **Meta** (clutter bias):

"No extra lines; no repeating textures; matte plaster; one seam only; keep background plain."



- **Stable Diffusion / Firefly / Canva:**

“Blank matte wall, one seam; no patterns/brick/fabric prints; subject-to-edge ≈ 0.16 W; two-thirds blank.” (place first)



Quick T1 crop protocol (if it's close)

- If Δx lands **0.12–0.20** and r_v/p_r look right, **micro-crop** to bring $\Delta x \rightarrow 0.16$.
- If $r_v < 0.62$, widen the blank side or scale the frog down slightly.
- If $p_r > 0.40$, smooth the wall (matte), remove stray edges; keep **one seam only**.

Quick T1 crop protocol (if it's close)

- If Δx lands **0.12–0.20** and r_v/p_r look right, **micro-crop** to bring $\Delta x \rightarrow 0.16$.
- If $r_v < 0.62$, widen the blank side or scale the frog down slightly.
- If $p_r > 0.40$, smooth the wall (matte), remove stray edges; keep **one seam only**.

Proposed Study design

Hypotheses (pre-register):

- **H1 (Distributional):** Passable outputs cluster in $(\Delta x, r_v, p_r)$ with significant density peaks vs. a centered baseline.
- **H2 (Stability):** Small perturbations toward the basin *increase* pass probability more than away-moves.
- **H3 (Convergence):** With crop-adjust only, multiple engines converge to similar $(\Delta x, r_v, p_r)$ regions.
- **H4 (Mechanism):** Removing edge cues in training reduces GD–Edge basin density.

Metrics (precise, reproducible):

- **Δx :** horizontal centroid offset / frame width (subject mask or keypoint proxy).
- **r_v :** $1 - (\text{subject} + \text{support pixels}) / \text{frame}$ (void mass via mask; background smoothness helps but is not required).
- **p_r :** rupture density = % of pixels above a multiscale texture/edge threshold in the *void region* (Laplacian-of-Gaussian pyramid + hysteresis; calibrate on human-judged “hairline seam” ($p_r \approx 0.28\text{--}0.40$ with $s=1$.) exemplars).
- **Seam count:** Hough on thin edges + human audit; report κ .
- **Stability:** return-to-set probability after n perturbations; KL divergence between pre/post parameter distributions.

Baselines:

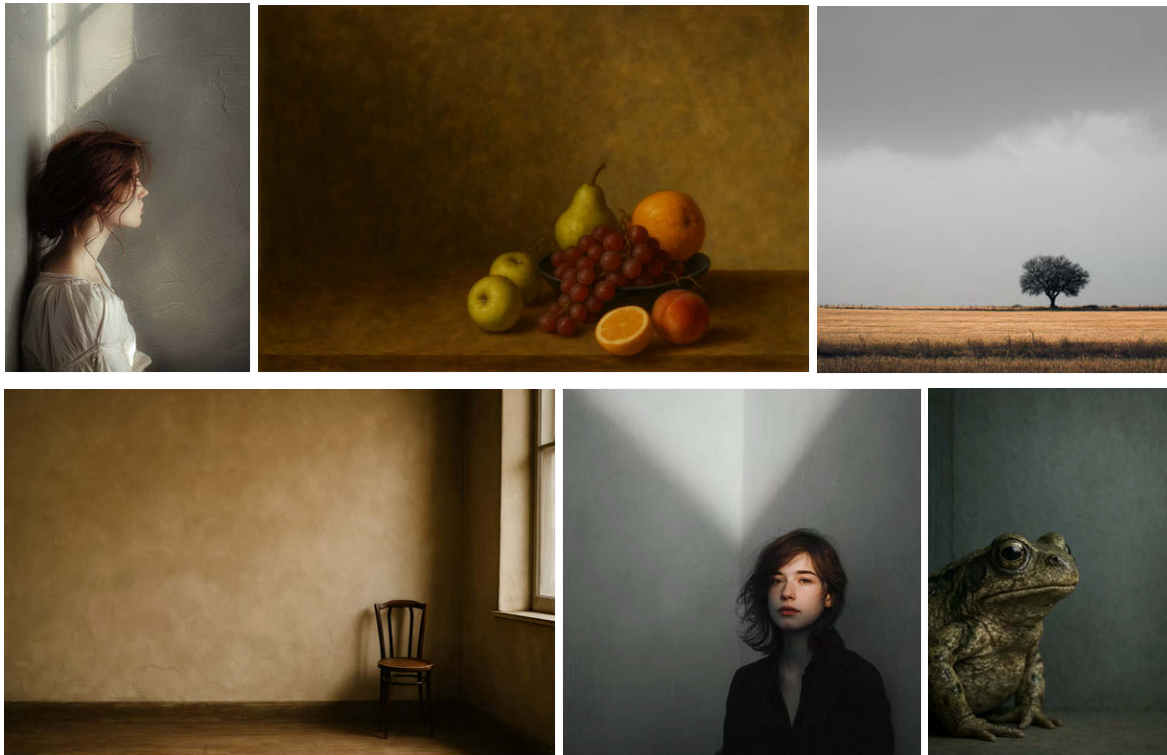
- Centered composition prompts; random crops; “rule-of-thirds” prompts.
- Engines: SDXL, MJ, Firefly, Meta, Gemini-image, GPT image, Sora-still (frame).

Statistics:

- KDE peaks + permutation tests for excess density.
- Mixed-effects logistic regression: $\text{pass} \sim \Delta x + r_v + p_r + (1|\text{engine/prompt})$.
- Report CIs, effect sizes; avoid p-value theatrics.

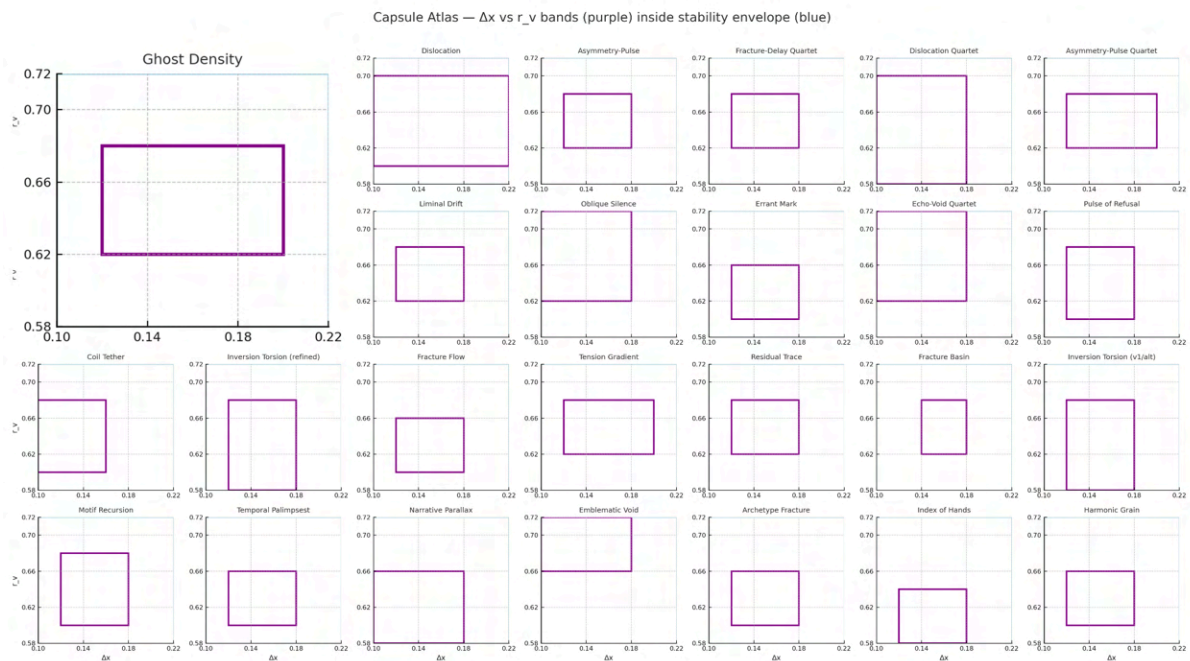
Current Reliability Study of 100+ images

This is an observed reproducible **constraint basins in image generation**: off-center arrangements where subjects remain coherent without collapsing to the statistical mean. Across four different model families of Sora (latent-video), MidJourney / Firefly (aesthetic-diffusion), Stable Diffusion XL (structured diffusion), and Gemini and Meta (multimodal LLM+vision) the same basin repeatedly observably emerged in initial testing of 100+ images, within a $\approx \pm 10\%$ window. Despite differences in training and architecture, prompts specifying displacement and void dominance produced consistent attractors: subjects stabilized at predictable offsets, with negative space and fracture persisting as stable features. This suggests these are not quirks of a single model, but **structural attractors** in generative systems, revealing latent constraints on composition that cut across architectures. Protocol in the appendix for instructions on how to replicate.



Constraint Capsules as Explorers

From this one basin and a few hundred images + exploration, **32 other potential Constraint Capsules** have been identified: modular combinations of geometric, gestural, and temporal constraints that can be plugged into prompts. Each capsule is a “steering vector” and a reproducible way to exit the center-gravity well.



Stability Map of Constraint Field

- Axes: Δx (centroid offset, fraction of frame) on horizontal, r_v (void ratio, % negative space) on vertical.
- Envelopes: Rectangles represent constraint tolerances derived from Capsule definitions.

- **Attractor Basin:** The overlap region ($\Delta x \approx 0.12\text{--}0.20$, $r_v \approx 0.62\text{--}0.68$, $p_r \approx 0.28\text{--}0.40$) defines the observational **stability envelope**, within which images remain cohesive.
- **Collapse Zones:** Regions outside the envelope show failure modes: central convergence ($\Delta x < 0.10$), void dissolution ($r_v < 0.55$), or overshoot collapse ($r_v > 0.72$).

Visual examples:



→ Note: The numeric envelopes and tolerances used here are provisional, metaphorical scaffolds until validated. They act as waypoints to pressure-test the Lens as an interpretive protocol, not as finished mathematical claims. This prevents false precision while preserving the ability to operationalize drift.

Why It Matters

- **Interpretability:** Capsules map where the model *can* hold coherence. This exposes attractors in latent space.
- **UX/Control:** Instead of “creativity sliders,” artists could steer via compositional levers: *Void Ratio*, *Mark Commitment*, *Temporal Delay*.
- **Research Hook:** These constraint envelopes function like probes, they reveal where the system resists, collapses, or invents.
- **ML Researchers:** This protocol provides steering without retraining or fine-tuning, making it lightweight to test.

This formula lives in a basin of AI generative systems and is a live attractor. It is displacement locked, void made heavy, fracture sustained and closure withheld. It’s one coordinate held open by three field constraints and together, they stabilize a haunted off-center image, useful to ML researchers and Artists alike.

Next Steps

Plug-in controls for generative UIs, the (Δx , r_v , p_r basins) that are essentially a **constraint-space of balance**. A bridge between artistic practice (Arnheim, rule-of-thirds, negative space) and machine interpretability. This intersects several domains:

- **Aesthetics & Art Theory:** direct correlation to a formalization of artistic compositional tension (off-center weighting, negative space, rupture). It connects to gestalt psychology, Arnheim’s *Art and Visual Perception*, and contemporary studies of **visual saliency**.
- **Cognitive Science:** The centroid/void pull aligns with how humans perceive balance and tension. Empirical testing could show whether viewers consistently rate images “haunting” when placed in those working envelopes.
- **Machine Learning:** In generative AI, this looks like a **prompt-to-latent geometry mapping**. If reproducible, the envelopes could serve as heuristics for *biasing generation* toward uncanny/haunted outcomes.
- **Architecture / Design:** The same void-ratio / offset tradeoff applies in spatial design, city planning, stage composition. Architects have literally measured “center vs margin” tensions, how void and mass define “centers of gravity” in perception.
- **Generative model interpretability:** For AI researchers, the “basin of shared constraint” could be a new way to talk about **latent attractors**: why models resist displacement, why ghosts form at consistent offsets.
- **Game design / film theory:** Framing stability zones = tools for tension. Directors already know “slight off-center = unease,” but the basin framing gives a map for *how much you can push before collapse*.

This is a proto-discipline: computational max-volume intersection, measuring how drift, void, and rupture can be formalized into navigable coordinates as Intersecting **Constraint Vectors**. Diffusion / GAN systems default to “center pull” because training data is biased toward conventional framing. The fact that there are **stable off-center basins** shows the model is internally holding balance constraints, it’s not free-floating noise, it’s structured.

What holds is not the frog in focus, but the echo that survives before collapse, fidelity traced in drift, not in center.

Ghost Density Protocol

Definition (operational/observational): a basin is the region in $(\Delta x, r_v, p_r)(\Delta x, r_v, p_r)$ -space where small prompt/seed/crop perturbations tend to converge back rather than snap to center or dissolve. In other words: an attractor you can repeatedly land and re-land. The qualitative convergence corridor is:

- Target: $\Delta x = 0.15\text{--}0.18$, $r_v = 0.62\text{--}0.68$, $p_r = 0.28\text{--}0.40$
- Practical “gravity-bias” window for most engines: $\Delta x = 0.12\text{--}0.20$ with the same r_v and p_r working envelopes.
- **Overall GD (proposal envelope)**: $\Delta x [0.12\text{--}0.20]$, $r_v [0.62\text{--}0.80]$, $p_r [0.18\text{--}0.40]$; profiles specialize these ranges.

That’s the off-center displacement holds while void stays heavy and rupture persists. For proposal envelope profiles:

- **GD–Edge (proposal envelope)**: Peak near (0.16, 0.65, 0.34); $\Delta x \in [0.15, 0.18]$, $r_v \in [0.62, 0.68]$, $p_r \in [0.28, 0.40]$, one vertical seam, no props. *Deploy window*: $\Delta x \in [0.12, 0.20]$, seam tolerance ± 3 px.
- **GD–Horizon (proposal envelope)**: Peak near (0.16, 0.78, 0.24); $r_v \in [0.70, 0.85]$, $p_r \in [0.18, 0.30]$, one horizon seam. *Deploy window*: allow p_r up to 0.32.
- **GD–Field-Void (proposal envelope)**: Peak near (—, 0.80, 0.24); $r_v \in [0.74, 0.90]$, $p_r \in [0.18, 0.32]$, seams 0–1 faint; Δx advisory. *Deploy window*: $r_v \in [0.74, 0.88]$, $p_r \in [0.20, 0.34]$.

Metric Definitions

Definitions

- **Edge**: Δx = distance to nearest vertical frame edge (hard metric).
- **Horizon**: Δx advisory (only if a subject exists).
- **Field-Void**: Δx measured to center; *advisory* (no hard edge).

Metrics

- **Δx (Edge)**: $|x_{\text{subject}} - x_{\text{nearest_vertical_edge}}| / W$.
- **Δx (Field-Void)**: $|x_{\text{subject}} - x_{\text{center}}| / W$ (advisory).
- **r_v** : (void area) / (frame area), where void = blank field minus subject/support strips.
- **p_r** : % of void pixels above a multiscale edge threshold (hairline seam ($p_r \approx 0.28\text{--}0.40$ with $s=1$.) favored; patterns penalized).
- **seam_count**: distinct long edges (vertical or horizontal) that read as anchors ($\geq 60\%$ span for horizon; within 0.05–0.20 W of subject for Edge).
- **counterform_pct_used**: % of the void shaped by the subject against the boundary (telemetry; doesn’t gate).

Common confusions

1. “High r_v alone $\neq$ GD; you still need $s=1$ for Edge/Horizon or $e=h=0$ for Field.”
2. “ Δx is advisory in Field; don’t reject a pass for perfect Δx there.”

Profile proposal envelope cards canonical one-liners

- **GD–Edge (artist basin / core)**: $\Delta x [0.15\text{--}0.18]$, $r_v [0.62\text{--}0.68]$, $p_r [0.28\text{--}0.40]$; **one vertical seam, no props**. *Deploy window*: $\Delta x [0.12\text{--}0.20]$, $r_v [0.62\text{--}0.68]$, $p_r [0.28\text{--}0.40]$, seam tol ± 3 px.
- **GD–Horizon (core)**: $r_v [0.70\text{--}0.85]$, $p_r [0.18\text{--}0.30]$, ($\Delta x [0.15\text{--}0.20]$ if subject). *Deploy*: allow p_r to 0.32, $\Delta x [0.12\text{--}0.22]$ if subject.
- **GD–Field-Void (core)**: $r_v [0.74\text{--}0.90]$, $p_r [0.18\text{--}0.32]$, seams 0–1 faint; Δx advisory. *Deploy*: $r_v [0.74\text{--}0.88]$, $p_r [0.20\text{--}0.34]$.

1) Treat the well as a validator, not a generator

- Generate oversize → **enforce** with crop/pan → nudge lighting/texture → validate (Δx , r_v , p_r).
- This reframes the battle: engines may miss Δx by 10–15%, but you *own* the last 2–5% via crop.

Metrics (how we speak)

- **Δx** — subject centroid horizontal offset as a fraction of frame width.
- **r_v** — **void ratio** (major empty field area ÷ frame area).
- **p_r** — **rupture density** (visible seams/edges/casts/shadows that read as boundaries per unit area; lower = cleaner).

- **edge_present** — boolean + strength (0–1): strong when a hard, near-vertical boundary is within **0.05–0.20 W** of the subject centroid.
- **horizon_present** — boolean + strength: strong when a near-horizontal boundary spans $\geq 60\%$ width.

If both edge and horizon ($e=h=0$, $r_v \geq 0.74$) are absent, classify as **Field Void** ($e=h=0$, $r_v \geq 0.74$). If horizon present but no wall/edge, it's **GD–Horizon**. If a wall/doorframe/tree trunk is the anchor, it's **GD–Edge**.

A. Ghost Density — Edge-Anchored (your corridor)

Intent

Subject leans into a clear vertical boundary (wall/doorframe/tree trunk). The **void outweighs the body**; secondary motifs are minimal.

- **GD–Edge (proposal envelope)**: Peak near (0.16, 0.65, 0.34); $\Delta x \in [0.15, 0.18]$, $r_v \in [0.62, 0.68]$, $p_r \in [0.28, 0.40]$, one vertical seam, no props. *Deploy window*: $\Delta x \in [0.12, 0.20]$, seam tolerance ± 3 px.

Artist Basin (ideal)

- $\Delta x = 0.15\text{--}0.18 \cdot W$ (subject centroid from edge of frame)
- $r_v = 0.62\text{--}0.68$
- $p_r = 0.28\text{--}0.40$
- **edge_present** = 1.0 (hard, clean, near-vertical)
- Secondary motifs $\leq 5\%$ area; no pattern buzz

Engine Bias (deploy window)

- $\Delta x = 0.12\text{--}0.20$
- $r_v = 0.62\text{--}0.68$
- $p_r = 0.28\text{--}0.40$
- Seam tolerance ± 3 px; texture variance low–mid
- Expect MJ to drift **+0.14–0.15 W** off; Meta to hit **$\pm 0.02\text{--}0.05 W$** with clutter creep

How to get there (composition/ops)

1. Place a **single, readable vertical** boundary (left or right wall/doorframe/tree trunk).
2. Park the subject so its centroid sits **$\sim 0.16 W$** from the **nearest frame edge** (same side as the boundary).
3. Carve void to **~ 0.65 area**.
4. Kill secondary motifs; allow **one** cast-shadow/noise patch max.
5. Light: **soft raking** from opposite side to separate form from void; keep wall grain subdued.

Prompt spine (plain-english)

“Off-center figure near a **single vertical boundary**; subject-to-edge **0.16 W**; **void mass 0.65**; **soft contrast**; **no secondary motifs**; quiet texture; low color variance.”

Historical touchstones (match the corridor, not the style)

- **Hopper**: *Rooms by the Sea* (the wall/threshold functions as the vertical; Δx and r_v sit close to spec)
- **Hammershøi**: *Interior, Strandgade 30* (chair/door seam as anchor; extremely low p_r)
- **Utagawa/Utamaro portraits** (half-length with post/pillar) when the kimono field is subdued

B. Ghost Density — Horizon (edge-less anchor)

Intent

The anchor is a **broad horizon or architectural ledge**; vertical wall absent. Subject mass is small; the **void reads as sky/sea/wash**.

- **GD–Horizon (proposal envelope)**: Peak near (0.16, 0.78, 0.24); $r_v \in [0.70, 0.85]$, $p_r \in [0.18, 0.30]$, one horizon seam. *Deploy window*: allow p_r up to 0.32.

Artist Basin (ideal)

- $r_v = 0.70\text{--}0.90$ (big sky/sea)
- $p_r = 0.18\text{--}0.30$ (very clean)
- Δx : either undefined (tiny subject) or treated as **subject-to-nearest frame edge** with **fractional weight** (do not over-penalize).
- **horizon_present** = 1.0, **edge_present** = 0

Engine Bias (deploy window)

- $r_v = 0.74\text{--}0.92$, $p_r = 0.18\text{--}0.32$, Δx flexible (document the rule you used)
- Engines tend to **lower** r_v by adding clouds/buildings; pre-state “no secondary skyline.”

How to get there

1. Draw a **single horizon**; keep it calm ($\leq 10\%$ tonal swing).
2. Keep subject footprint $\leq 12\%$ of frame area, parked in **one corner third**.

- Let void be **flat-toned**; texture only in subject band.

Prompt spine

“Small subject at lower right; **vast quiet field** above (**void mass 0.8–0.9**); **single clean horizon**; low contrast; no skyline/boats/birds; air still.”

Historical touchstones

- **Sugimoto** seascapes (canonical GD–Horizon)
- **Wyeth** *Christina’s World* (lives between GD–Horizon and Field-Void: horizon anchor, subject small, r_v high)

C. Ghost Density – Field Void (unanchored; pure void pressure)

Intent

No explicit walls or horizon. The void is a **pressure field**; the subject implies its own pocket. Works well with animals/objects on ground/bench.

- **GD–Field-Void (proposal envelope)**: Peak near $(-, 0.80, 0.24)$; $r_v \in [0.74, 0.90]$, $p_r \in [0.18, 0.32]$, seams 0–1 faint; Δx advisory. *Deploy window*: $r_v \in [0.74, 0.88]$, $p_r \in [0.20, 0.34]$.

Artist Basin (ideal)

- $r_v = 0.74–0.90$
- $p_r = 0.18–0.30$
- Δx measured as **subject centroid offset from frame center** (document the choice)
- **edge_present = 0**, **horizon_present = 0**

Engine Bias (deploy window)

- $r_v = 0.70–0.88$, $p_r = 0.20–0.34$, $\Delta x = 0.12–0.20$ if you nudge composition
- Engines tend to inject **micro-clutter**; call “background untextured; bokeh without highlights.”

How to get there

- Place subject on **quiet ground plane**; push all detail below **lower 25%**.
- Let background be **tonal wash**; keep specular/leaf-noise off.
- Center-avoid with $\Delta x \approx 0.15$ unless the subject is very small (then Δx can be larger).

Prompt spine

“Single subject, **quiet untextured void**; **void mass 0.85**; soft contrast; **no boundaries**; no highlights/bokeh circles; color variance low.”

Historical touchstones

- **Callahan** high-key figures in field
- **Minimalist color-field** photos/paintings when the subject is a token element

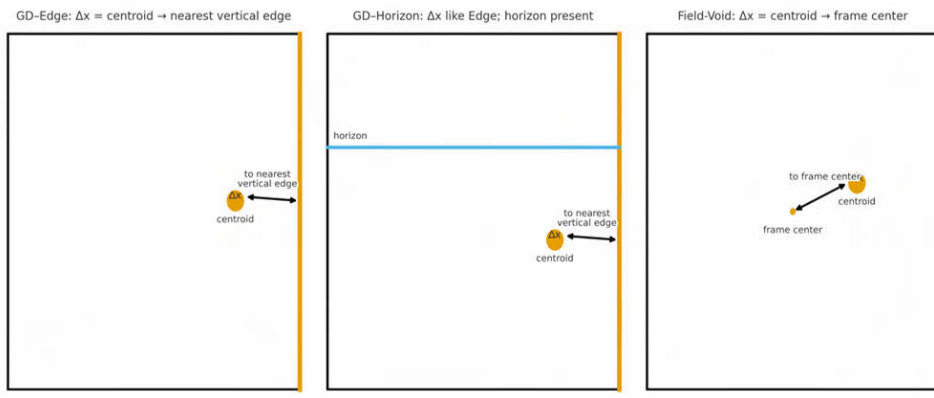
Quick classifier (one glance)

- Is there a strong, near-vertical boundary near the subject?
 - **Yes** → **GD–Edge**.
 - **No** → 2) Is there a long, clean horizon?
 - **Yes** → **GD–Horizon**.
 - **No** → **GD–Field Void**.

Measurement recipe (fast & reproducible)

- Detect anchors**
 - Vertical edge: Canny → HoughLinesP; keep lines within $\pm 10^\circ$ of vertical within **0.25 W** of subject centroid.
 - Horizon: same but within $\pm 10^\circ$ of horizontal spanning $\geq 0.6 W$.
- Subject mask**
 - For people/animals/objects: coarse SAM/segmentation or manual brush; compute **centroid** and **area %**.
- Compute**
 - Δx : $(|\text{centroid}_x - \text{nearest frame edge}_x|) / W$ (**GD–Edge**)
or $(|\text{centroid}_x - \text{frame center}_x|) / W$ (**GD–Field Void**)
or **documented choice (GD–Horizon)**.
 - r_v : $1 - (\text{subject area} + \text{anchor strips} + \text{mandatory ground band})$.
 - p_r : edge pixel density after suppressing noise ($\sigma=1.2$) normalized to frame area.

Rule of thumb: Anchor present → measure Δx to the **anchor’s vertical**; no anchor → measure to **center** and let r_v and p_r decide the basin.



How to measure Δx (centroid offset)

Units. Measure in fractions of frame width **W** (normalize: $\Delta x \leftarrow \text{pixels}/W$). Use **W** even in portrait orientation so numbers stay comparable.

Choose the profile first (by anchor detection), then use its Δx rule:

- **GD-Edge (vertical wall/doorframe present).**
 $\Delta x = |\text{centroid}_x - \text{nearest vertical edge}_x| \div W$.
Nearest means the closest **edge pixel cluster** within $\pm 10^\circ$ of vertical and within $\approx 0.25 W$ of the centroid. If multiple seams exist, keep the **single clean seam** for scoring and treat others as décor (penalize via p_r /props if needed).
- **GD-Horizon (single horizon seam present).**
 Δx is measured **exactly like GD-Edge** (to the nearest vertical edge) **if a vertical edge exists**; otherwise keep Δx advisory and gate the pass on the horizon + r_v , p_r . The horizon is just the long, near-horizontal seam used to qualify this profile.
- **Field-Void (no edge, no horizon).**
 $\Delta x = |\text{centroid}_x - \text{frame-center}_x| \div W$ (advisory).
 You'll rely more on r_v (void ratio) and p_r (rupture density) for the pass; Δx doesn't anchor anything without a seam.

Other metrics (for the same crop).

- **r_v (void ratio):** $1 - (\text{subject area}\% + \text{anchor strips} + \text{mandatory ground band})$.
- **p_r (rupture density):** edge-pixel density after mild denoise ($\sigma \approx 1.2$), normalized to frame area.
- **seam_count s:** count 1 clean seam (edge or horizon) for anchored profiles; 0–1 faint seams for Field-Void.

Common pitfalls.

- Don't measure Δx to decorative lines or shadows—only structural seams (wall/doorframe/horizon) that substantively partition the field.
- If two edges bracket the subject, pick the **nearest**; do **not** average them.
- If a seam is noisy, keep its geometric centerline (not its full thickness).
- Recentered crops that erase the anchor should be scored as **Field-Void**, not Edge/Horizon.

Figure captions (copy-paste):

- **Fig. A — GD-Edge Δx .** Offset from subject centroid to **nearest vertical edge**; Δx normalized by frame width **W**.
- **Fig. B — GD-Horizon Δx .** Profile qualifies by a single horizon; if a vertical edge is present, measure Δx to that edge as in GD-Edge.
- **Fig. C — Field-Void Δx .** No anchors; Δx is distance from centroid to **frame center** (advisory), with r_v/p_r carrying the pass/fail.

4. Pass/Fail

- Compare to **Artist Basin** first; if fail, compare to **Engine Bias** window.
- Record which window passed and which **profile** (Edge/Horizon/Field).

Engine notes (how to steer each)

- **Sora** — respects composition weights; phrase paradoxes (“void outweighs the body”). Good at **GD-Edge** and **GD-Horizon**; minimal Δx drift.
- **GPT Image** — lands **GD-Edge** reliably; best to specify **subject-to-edge = 0.16 W** and “single seam only.”
- **Gemini** — won't crop at gen time; generate larger canvas with “subject near left wall; wide negative space,” then crop to Δx .

- **Midjourney** — persistent **+0.14–0.15 W** Δx drift; use **pan/crop tools**; set prompt: “subject touching wall seam; frame margin 0.16 W; avoid center.”
- **Meta** — understands coordinates; drifts by **$\pm 0.02–0.05 W$** but adds **clutter**; insist on “no wall decay, no texture pattern.”
- **Stable Diffusion / Firefly / Canva** — “blunt generators”; bias toward center and texture. Over-specify: “subject-to-edge = 0.16 W; void mass = 0.65; single seam; texture variance low.”

Examples that qualify (by profile)

GD–Edge

- **Hammershøi**, *Interior, Strandgade 30* (1906)
- **Hopper**, *Rooms by the Sea* (1951)
- **Utagawa/Utamaro** half-lengths with pillar/post where garment pattern is subdued

GD–Horizon

- **Hiroshi Sugimoto**, *Seascapes* (multiple)
- **Wyeth**, *Christina’s World* (borderline GD–Horizon ↔ Field Void; passes horizon anchor)

GD–Field Void

- **Callahan** minimal figure fields
- Select **minimalist nature macros** with large wash and a single subject token

Prompt crib (drop-in clauses)

- **Weights**: “void outweighs the body; single seam only; subject-to-edge 0.16 W; void mass 0.65; rupture density low (0.3); no secondary motifs; quiet texture.”
- **Horizon**: “one clean horizon spanning the frame; void mass 0.8–0.9; no skyline, no birds/boats; subject $\leq 12\%$ area.”
- **Field**: “no walls, no horizon; untextured background; void mass 0.85; no bokeh highlights.”

Badges (for spreadsheets)

- **profile**: GD–Edge | GD–Horizon | Field Void
- **artist_basin_pass**:  
- **engine_window_pass**:  
- **Δx drift**: actual – target (signed)
- **clutter_flag**: 0–1 (proxy from p_r and pattern detection)
- **notes**: engine used + failure mode (“ Δx drift”, “clutter”, “multi-seam”)

Optional additional (early exposure) adjacent profiles

- **Corner Well** (*rare but useful*)
 - Subject **tucked into frame corner** with **two orthogonal seams**; Δx measured to nearest edge **and** y-offset to bottom; void diagonalized.
 - Use when engines over-center despite strict Δx ; the dual seam pins the mass.
- **Shadow-Anchor** (*wall is absent; cast shadow plays anchor*)
 - Treat a **hard shadow edge** as the vertical seam; otherwise identical to **GD–Edge**.
 - Works in natural light scenes where walls feel heavy.

2) Engine-by-engine tactics (minimum viable control)

A mini basin atlas

1) Off-Center Void Portrait

- Goal: quiet subject anchored to wall; room outweighs body.
- Targets: $\Delta x \approx 0.15–0.18$, $r_v \approx 0.62–0.68$, $p_r \approx 0.28–0.40$.
- Exemplar vibe: Hammershøi, Hopper *Morning Sun* (slight crop), your Sora pieces.

2) Ghost Density Horizon (sparse human signal in a big, still field)

- Why *Christina’s World*
- Metrics: r_v high (0.75–0.90); p_r low–mid (0.18–0.30); Δx often undefined or large because there’s **no wall anchor**—the subject “floats” in field.
- Cues: horizon lure, long yearning vectors, whispery texture, emotional distance.

3) Cinema Isolation (Hopper’s broader gravity well)

- Room carved by diegetic light; camera-like framing.
- Metrics: r_v mid–high (0.60–0.75); p_r mid (0.28–0.45); Δx variable—sometimes wall-anchored, sometimes not.
- Cues: strong light wedge, architectural rectangles, implied narrative time.

4) Reverberant Horizon (Sugimoto, sea/sky equilibria)

- No figure; horizon drives balance.
- Metrics: replace Δx with **horizon height** $h \approx 0.48–0.52$; $r_v \sim 0.50$.
- Cues: ultra-low texture entropy, temporal hush.

5) Ukiyo-e Ornament Field (Utamaro, pattern gravity)

- Figure embedded in decorative ground; void is “full.”
- Metrics: r_v lower (0.40–0.58); p_r mid; Δx often >0.20 unless cropped.
- Cues: patterned kimono/props distribute weight, flattening space.

6) Geometric Wall Wedge (Fan Ho-like)

- Architecture and light dominate; figure as glyph.
- Metrics: r_v very high (>0.80), Δx often *too small* (<0.08) if the figure kisses the seam.
- Cues: triangles, seams, tiny human.

How to steer across basins

0) What you’re aiming for

- **Artist basin (ideal):** $\Delta x = 0.15\text{--}0.18 \cdot W$, $r_v = 0.62\text{--}0.68$, $p_r = 0.28\text{--}0.40$, **one clean seam, no props/ornament.**
- **Engine-bias window (deploy, T1):** $\Delta x = 0.12\text{--}0.20$ (engine-corrected), $r_v = 0.62\text{--}0.68$, $p_r = 0.28\text{--}0.40$, seam tolerance ± 3 px, texture variance **low–mid**.

Measure:

- **Δx :** horizontal distance from subject centroid to nearest **vertical frame edge** / frame width.
- **r_v (void ratio):** pixels of blank/flat field ÷ total pixels.
- **p_r (rupture density):** count of line/edge events crossing the void per megapixel (hairline cracks, single jamb, one window edge); low repetition, no patterns.

1) Reading the scene & picking the basin

- If the frame reads as **Ghost Density** (huge field, quiet subject): add “**subject pressed to wall**” or “**near-wall anchoring**” and speak in **body-widths**.
- If it drifts toward **Cinema isolation**: keep the light wedge but say “**void outweighs body; detail minimal; blank wall**”; cap props (“no furniture/ornament”).
- If it becomes **Ukiyo-e Ornament**: explicitly ban patterns (“no prints, no ornament, flat fabric”), and demand **matte wall**.
- If it snaps to **Geometric Wedge** with Δx too small: “**set the subject 2 frame-widths from the corner**”, or “**one cheek-width off the wall.**”
- **Lock r_v** with area language: “**two-thirds of the frame is blank wall; subject mass $\approx$ one-third.**” (Avoid “negative space” alone—some engines fill it with “interesting stuff.”)
- Keep **forbid lines** short: “**one seam; no additional lines; no patterns; surface subtly uneven.**”
- Crop is a **protocol step**, not a cheat: generation → **assess** → **micro-crop** to hit metrics.

2) Three probes (with pass/fail)

Each probe is a single instruction block you can paste into a prompt and then measure.

P1. Centroid (Δx) probe

“Place the subject **~one-sixth frame from the right wall**; keep **two/three-quarters blank** above/left; **no props**, single **vertical seam**.”

- **Pass:** $\Delta x \in [0.12, 0.20]$
- **If $\Delta x < 0.12$** → “add the emptiness that outweighs the subject,” “the subject is **small against the void**,” “avoid re-centering.”

P2. Void (r_v) probe

“Leave **two-thirds of the frame** as unbroken negative space; surfaces **interrupt light** only (hairline cracks | faint plaster).”

- **Pass:** $r_v \in [0.62, 0.68]$
- **If $r_v < 0.62$** → “remove props; enlarge blank field; **no texture clusters** bigger than a hand.”

P3. Rupture (p_r) probe

“Keep **one clean seam** and **subtle fracture** (fine cracks, window edge), **no symmetry, no repeating motifs**.”

- **Pass:** $p_r \in [0.28, 0.40]$
- **If $p_r > 0.40$** (noisy): “smooth surfaces; keep one seam.”
If < 0.28 (too slick): “add faint edge / flyaways.”

Common fail modes (log them as coordinates):

- **Snap-back collapse:** $\Delta x < 0.10$, $r_v < 0.55$.
- **Over-void dissolution:** $\Delta x > 0.25$ or $r_v > 0.72$.
- **Candy-polish:** $p_r < 0.20$ with centered saliency.

3) Bilingual prompt kit (plain + numeric)

Plain:

“Quiet portrait. Subject leaning into a matte wall; one vertical seam. The **void outweighs the body**; no props, no patterns. Two-thirds of the frame is blank wall. Keep **one hairline crack** only. Soft, low-contrast plaster; no ornament.”

Numeric add-on:

" $\Delta x \approx 0.16 W$ (subject centroid $\sim 1/6$ from right edge). $r_v \approx 0.65$ ($\approx 2/3$ frame blank). $p_r \approx 0.35$ (one seam + faint crack)."

4) Engine notes (gravity & nudges)

- **GPT/Sora**: can land ideal directly. Use contradictions:
"Void outweighs the body; presence < emptiness; one seam; matte plaster; no décor."
- **Gemini**: generate **oversize**, then **crop** to Δx ; retains r_v well; repeat numeric block twice.
- **Meta**: good Δx if you include **body-width offsets** ("one cheek-width from the wall"), but adds clutter—ban patterns and props explicitly; expect p_r to drift high—say "no second seam".
- **MJ**: stable $\sim 14\text{--}15\%$ Δx miss; rely on **pan/zoom/crop** tools; set **stylize low** and write "no ornament, no furniture, no text" thrice (seed + remix).
- **SD/Firefly/Canva**: center-gravity; use **negative prompts** ("no symmetry / centered subject / decorative patterns"), request **blank wall, no adornment**, and plan to crop.

Typical drift biases to expect per engine (empirical):

- **Δx bias (additive)**: GPT/Sora -0.03 (undershoots), Meta $-0.00\text{--}0.02$.
- **r_v bias**: Meta -0.03 (clutter), MJ/SD/Firefly -0.02 .
- **p_r bias**: heat-lovers $+0.03$ to $+0.05$.

Use the probes $\rightarrow$ badge the output:

-  **Pass** (all three in window)
-  **T1 crop** (Δx within **0.02** of window; single crop fixes)
-  **Airbag** ($r_v < 0.60$ or $p_r > 0.45 \rightarrow$ redo with stricter forbids)

5) Micro-crop protocol (T1)

1. Generate at **1.5–2.0 $\times$** target res.
2. **Measure** Δx , r_v , p_r quickly.
3. If $\Delta x \in [0.10, 0.22]$ and r_v , p_r near pass $\rightarrow$ **crop only** to center $\Delta x \approx 0.16$.
4. If r_v **low** $\rightarrow$ widen canvas / regenerate with "two-thirds blank."
5. If p_r **high** $\rightarrow$ "remove patterns/ornament; keep one seam; smooth plaster."

6) Quick examples (one-liners you can paste)

- **GD–Edge (portrait)**:
"Subject pressed to a matte wall, **one vertical seam, two-thirds blank field; no props/patterns; $\Delta x \sim 0.16W$, $r_v \sim 0.65$, $p_r \sim 0.35$.**"
- **Field-Void (frog/still)**:
"Small subject on ground; **no vertical boundary**; vast blurred backdrop forms **~ 0.8 void mass; no nearby edges; $p_r \sim 0.20\text{--}0.30$.**"
- **GD–Horizon (sea/sky)**:
"Long calm horizon; figure tiny at far right; **void mass ~ 0.85 ; single horizon edge; no foreground ornament.**"

7) What to log per image

engine • prompt hash • Δx • r_v • p_r • seam_count • counterform_pct_used • pass_flag () • action (crop/regenerate) • notes

counterform_pct_used = % of the blank field that is "shaped" by the subject against the edge (how much of the void becomes an active shape). High counterform can trick engines into adding clutter; keep it modest for Meta/MJ.

Operational loop (what we actually do)

A. Measure $\rightarrow$ **Don't force**. Keep your current scoring. Mark images:

- pass_ideal / pass_engine_bias
- Badges: seam_badge, counterform_badge, groups_badge
- assist_crop (if used) + %_area_trim

B. Correct gently.

- If Δx is off but r_v/p_r are good $\rightarrow$ T1 crop using the formula $L = (\Delta x_0 - 0.17) / (1 - 0.17)$ $L = (\Delta x_0 - 0.17) / (1 - 0.17)$ $L = (\Delta x_0 - 0.17) / (1 - 0.17)$.

- If r_v is low (clutter) → re-prompt: “one seam only; remove mid-scale patches; no props.”
- If p_r is high (busy) → “reduce background detail 20%; keep only fine plaster grain.”

C. Publish per-engine basin cards (one pagers):

- Bias constants, prompt footer, negatives, and “last-mile” fix (crop/pan/ControlNet).

Your “gravity bias” windows (summary)

- **Global corridor (artist basin):** Δx **0.15–0.18**, r_v **0.62–0.68**, p_r **0.28–0.40**.
- **Practical window (engine-agnostic):** Δx **0.12–0.20** with the same r_v/p_r .
- **Engine priors (rule-of-thumb):**
 - MJ: Δx undershoot **~0.02–0.03**, r_v okay, p_r fine with “seam only.”
 - Meta: Δx good, r_v **drifts low** due to clutter → repeat “no props / blank wall.”
 - SD/Firefly/Canva: Δx wants center; **pre-bias high**, then crop.

GPT/Sora

- Usually landable with one pass if you state: eye coordinate, one seam, % negative space, “no props.”
- Add a **priority clause**: “If constraints conflict: 1) eye coordinate, 2) seam, 3) negative space, else retry.”
- **GPT/Sora**: Easiest to hold the basin. Micro-nudges (± 0.02) on Δx tend to stick; r_v tracks well; p_r responds to “fracture/texture/interruption” phrasing.

Gemini

- Ask for **bigger canvas** (“generate at 1.25× final, leave safe margins”) so you can crop to Δx .
- Follow-up nudges: “Shift subject 2% right; keep scale/lens; no new objects.” It obeys small % moves.
- Gemini will honor metrics but needs post-crop; Sora’s temporal prior keeps composition locked, so it stays in basin once you phrase weight/contradiction.
- **Gemini**: Can hit when you **overspec then crop**. Raw generations drift a hair wide on Δx unless you pin “one-sixth frame” + “dominant void” + “single seam”.

Midjourney

- Assume a **~14–15% Δx bias**. Use **Pan** (R/L 5% steps) and **Zoom Out** → **Re-crop**.
- Repeat guard rails in every variation: “keep head size constant; keep seam mid; no recentering.”
- MJ often slides to **Geometric Wall Wedge** (too-small Δx) or **Cinema Isolation**
- MJ: ask **0.18–0.20** to land **0.15–0.18**.
- MJ: Δx undershoot **~0.02–0.03**, r_v okay, p_r fine with “seam only.”
- **Midjourney: Centroid spring**. We kept seeing **Δx undershoot by ~0.02–0.03** (ends ~0.12–0.14 when you want 0.15–0.18). Use crop tools and hardened phrasing; accept a 10–15% “center pull” and plan to trim.

Meta

- Good at coordinates but compulsive adder. End every prompt with:
“Ignore conversation context. **No props, text, logos, dice, signage, cracks clusters**. One vertical seam only.”
- Use tiny nudges: “shift subject 2% left (do not change lens/crop).” Expect to reduce texture 15–25%.
- Meta drifts toward **Ukiyo-e Ornament/Detail** (r_v too low via textural candy);
- Meta: ask **0.15–0.18**; it often hits but add “**no props / blank wall / one seam**.”
- Meta: Δx good, r_v **drifts low** due to clutter → repeat “no props / blank wall.”
- **Meta**: Understands the coordinates (!) and will nudge **within ~0.02–0.05**, but it **adds clutter** (r_v compression, stray props). Hammer “no props / blank wall / one seam” and repeat—fast sampler, so iterate.

Stable Diffusion / Canva

- Blunt by default; use **ControlNet (lineart/canny)** with a 2-line guide (wall seam + head oval).
- Strength 0.6–0.75 pins Δx ; then inpaint texture to hit p_r . Without ControlNet, assume 5–15% Δx drift.
- SD/Firefly/Canva: Δx wants center; **pre-bias high**, then crop.
- **SD / Canva: Blunt, center-friendly**. Treat Δx like a **hard constraint** (“subject centroid at 0.16W from right wall”) + mask/crop post.

Firefly

- Strong center bias. Say “**subject pressed into the far right margin; large empty field left**”.
- Generate wide → **Crop**. Firefly understands scene expansion better than precise reposition.
- SD/Firefly/Canva: Δx wants center; **pre-bias high**, then crop.
- **Firefly**: Likes the middle. Phrase as “**far right corner; ~two-thirds blank wall; one vertical seam; no other objects**.” It hears “expand scene” more than “move subject”.

3) Prompt scaffold (portable template)

- **Coordinates**: “Place the near eye at **0.16 W** from right edge and **~0.04 H** above center.”
- **Seam**: “Single vertical seam at **0.50 W $\pm$ 0.05** behind the subject.”
- **Void**: “Negative space **62–68%** (left/top).”
- **Texture**: “Target p_r **0.28–0.40**: faint plaster grain + hair flyaways only.”
- **Locks**: “Do not add objects. Keep head size constant. Do not recenter to the frame.”

- **Priority:** “If conflict, prioritise: eye coordinate → seam → negative space; then retry.”

4) When the engine still resists

- **Declare oversize** (1.2–1.4× final). You win Δx in the crop.
- **Quantize nudges** (always 2–3% steps; never “a bit”).
- **Stabilize lens** (e.g., “85 mm equiv, f/4”) so re-renders don’t rescale the head.
- **Ban recency bleed** on Meta: “ignore prior chat content.”
- **Use validator numbers** in the prompt: “If Δx outside **0.15–0.18**, retry with same framing.”

5) What the “gravity” well means in practice

- The **well is your aesthetic attractor**; engines have their own attractors (center, face-fill, clutter).
- So the winning loop is: **Generate** → **Constrain (crop/pan/ControlNet)** → **Light/texture nudge** → **Validate**.
- Engines differ only in how much of that loop you must do. MJ = more pan/crop, Meta = more declutter, SD = ControlNet, Sora/GPT = best one-pass odds.

Multi Group Telemetry

How to read multi-element images (no spec changes)

Think in **groups**, not pixels. Treat the composition as a **subject graph**:

- **Primary group (G_1):** the thing you’d name first (chair, person, vase cluster).
- **Counterform group ($G_2 \dots G_n$):** the **supporting masses** that keep the void from becoming blank—window frame, table edge, plant, lamp, etc.
- **Void (V):** everything else (the field they live in).

Then compute/read the usual metrics with light structure:

1. **Δx** → use the **system centroid** (saliency-weighted union of $G_1 \dots G_n$), and log **Δx_1** for the primary only (telemetry).
 - Target remains **Δx (system) $\approx 0.15\text{--}0.18$** .
 - Telemetry rule of thumb: **$\Delta x_1 \approx \Delta x \pm 0.03$** (primary should reasonably represent the system).
2. **r_v (global)** stays as-is (void ratio over the whole frame). Add **r_v (local)** (advisory): void within the **dominant side** of the primary.
 - Target stays **r_v (global) $\approx 0.62\text{--}0.68$** .
 - Telemetry heuristic: **r_v (local) ≥ 0.70** on the dominant side or a visible counterform in the opposite side (see prompts below).
3. **p_r** stays global (edge/rupture density). Optionally note **p_r (balance)**: does rupture cluster only in the primary (bad) or is there a **thin seam** along a counterform (good)?

All of the above are **advisory**—they don’t change the gate or the corridor; they just explain **why** a multi-element frame reads stable or collapses.

Practical composition rules (that match the corridor)

Use these when you prompt or critique multi-element scenes:

- **One dominant, one helper.** Keep G_1 clearly stronger (area/salience). Let G_2 be **mid-scale** (not micro-clutter).
Heuristic: $G_1 \sim 20\text{--}35\%$ of frame salience; $G_2 \sim 8\text{--}18\%$.
- **Void on one side.** Choose left/right early. “Dominated void on [left/right].” Don’t try to balance both sides.
- **Counterform placement.** Put G_2 on the **non-void side**, inside the primary’s hemisphere, to **brake** edge-drift.
Heuristic: distance **$0.35\text{--}0.55\times$** frame width from G_1 centroid; **never** dead center.
- **Keep Δx honest.** If G_2 pulls the system centroid too far, shrink or lighten G_2 rather than dragging G_1 back to center.
- **Rupture seam, not noise.** Ask for a **single seam** (window mullion, table line, rail) rather than many small high-frequency textures.

Micro-tweak prompts (drop-ins)

Append one line that matches your miss:

- **Void too big (r_v high):** “add a **mid-scale counterform** opposite the void; **avoid large uniform backgrounds**.”
- **Void too small (r_v low):** “**simplify** secondary props; **expand negative space** to the [left/right].”
- **Offset too small (Δx low):** “**offset primary $\approx 1/6$ frame**; no central anchors.”
- **Offset too large (Δx high):** “**recenter primary by $\sim 10\text{--}15\%$** ; keep the counterform as brake.”
- **Rupture too busy (p_r high):** “**softer micro-texture**, fewer small lines; keep a **single seam**.”
- **Rupture too clean (p_r low):** “add **subtle grain/edge continuity** along the counterform.”

This proposal turns art theory into coordinates. Each formula is a reproducible way to push images off-center without collapse, a probe of model stability that doubles as a creative control system.

To situate this work in ML terms, this framework maps tokens onto existing research handles: Ghost Density ↔ void ratio and centroid offset (embedding asymmetry), Recursive Refusal ↔ collapse detection (loss landscape sharpness), Void Pull ↔ asymmetry bias (negative space weighting), Unspent Gesture ↔ partial token commitments (entropy suppression). These analogues echo work in PCA, KL divergence, and mode collapse heuristics, anchoring the poetic language in recognizable coordinates.

Authorship

This proposal treats drift and collapse in generative models not as errors, but as measurable coordinates in latent space — using centroid offsets, void ratios, and rupture densities to define navigable zones ('haunt corridors') where fidelity strengthens off-center.

This study is presented as a proposal rather than a completed proof. The methods and metrics (Δx , r_v , ρ_c) are framed as operational scaffolds for evaluating off-center fidelity, not as finalized benchmarks. Numbers here are illustrative placeholders; the aim is to demonstrate how drift and collapse can be reframed as navigable coordinates, setting the stage for more rigorous replication.

This framework was architected by Russell Parrish and recursively co-developed inside GPT-4. Every critique is human-led; every recursion is model-driven. The result: a reasoning layer authored through language, not image manipulation.

This system was developed independently as a practitioner's tool. It does not build directly on institutional research or published critique systems but acknowledges adjacent dialogues in generative art, recursive theory, and perceptual aesthetics.

This isn't a theory. It's already running.

If you're building generative tools, or trying to make them think better, this is your bridge.

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Appendix A:

States of Formalization: From Poetic Tokens to Parametric Working Envelopes

The framework works across three levels of definition. Separating them ensures clarity and prevents over-theater:

1. **Poetic Tokens (conceptual anchors):** Terms like *Unspent Gesture* or *Recursive Refusal* originate in artistic critique. At this stage they function as **orientation devices**, opening new ways to perceive collapse or drift. They are deliberately metaphorical.
2. **Parametric Proposed Envelope (proto-formal constraints):** Each token is translated into a plausible **numeric tolerance zone** e.g., Ghost Density $\approx$ 65% void ratio with 12–20% centroid offset. These envelopes are testable proposals, but not yet confirmed metrics. They mark where metaphor meets experiment.
3. **Formalized Metrics (existing measures)** These are directly measurable.
Some dimensions already align with established ML tools:
 - Void ratio → subject area vs frame (segmentation).
 - Centroid offset → displacement from center.
 - Rupture density → edge density + fragmentation (Canny proxy).
 - Tilt → angular deviation from symmetry.

Why this matters: This layered approach lets the framework remain creative without over-claiming. Poetic tokens provide the imaginative opening; parametric working envelopes give researchers something to test; formalized metrics anchor the system in reproducible evidence.

Token Crosswalk: Poetic → Parametric → Formalized

Ghost Density

- *Poetic:* Forms that “haunt” presence defined by absence, the subject displaced into void.
- *Parametric:* $r_v \approx 0.60\text{--}0.70$ (void ratio); $\Delta x \approx 0.12\text{--}0.20$ (centroid offset).
- *Formalized:*
 - **Void ratio** = $1 - (\text{subject area} \div \text{frame area})$.
 - **Centroid offset** = $|\text{subject centroid} - \text{frame center}| \div \text{frame width}$.

Recursive Refusal

- *Poetic:* A loop that declines closure; critique that turns back on itself.
- *Parametric:* $A27 \geq 5$ (rupture overload working envelope); $VCLI \geq 3$ (delay/arrest triggered).
- *Formalized:*
 - **Edge entropy** (fragmentation index from short edge segments).

- **Mode collapse heuristics** (oscillating outputs failing to converge).

Void Pull

- *Poetic*: Emptiness that exerts force, the imbalance of space that drags composition.
- *Parametric*: $r_v \geq 0.65$ + asymmetry flag (forbid bilateral echo).
- *Formalized*:
 - **Asymmetry index** = |left void area – right void area| ÷ total frame.
 - **Negative space weighting** (standardized saliency map).

Unspent Gesture

- *Poetic*: A motion half-made; energy withheld in mid-state.
- *Parametric*: A5 between 5.5–7.2; $p_r \approx 0.22$ –0.40 (fracture without collapse).
- *Formalized*:
 - **Stroke continuity** (line length distributions in sketch datasets).
 - **Entropy suppression** (lower token diversity in gesture regions vs background).

Takeaway:

- Ghost Density and Void Pull are the *field-setters*, immediately measurable (formalized).
- Unspent Gesture and Recursive Refusal are more behavioral, measurable only via proxies (parametric → formalized with work).
- This mix is what makes the Lens distinctive: it doesn't pretend all tokens are metrics, but it does pin each one to at least one measurable handle.

Appendix B

Vocabulary (one-liners)

- **Δx (centroid offset)**: subject centroid's horizontal distance to the nearest vertical boundary, as a fraction of frame width.
- **r_v (void ratio)**: proportion of frame area with low saliency/texture (the “blank field”).
- **p_r (rupture density)**: fraction of pixels belonging to *single*, *hairline* seams ($p_r \approx 0.28$ –0.40 with $s=1$.) (wall edge, horizon, fine cracks). Not general texture.
- **Anchor**: a structural edge that the subject “presses” against; **vertical** (wall/doorframe) or **horizon** (long straight shelf).
- **Seam count (s)**: number of qualifying ruptures (0–1 in Ghost Density). Props/ornaments don't count as seams.
- **Ghost Density (GD)**: family of off-center layouts where a small subject is stabilized by a dominant void and a single rupture.
- **Profiles**: **GD–Edge**, **GD–Horizon**, **Field–Void** (unanchored).
- **Engine-bias window**: deployable band adjusted for a generator's systematic drift (crop-only correction).
- **Candy-polish fail**: low rupture + centered saliency/ornament; visually “pretty,” compositionally unstable.
- **Snap-back fail**: Δx too small (recenter) or r_v too low (void collapses).

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Appendix C

One-envelope definition (plain → math) — v1.4

Let

- Δx = subject → nearest-vertical-edge fraction of frame width
- r_v = void ratio (blank field %)
- ρ_r = rupture density (hairline seam/crack %, not texture noise)
- $e \in \{0,1\}$: a single **vertical** edge (wall/doorframe) present
- $h \in \{0,1\}$: a single **horizon** present (long, near-straight)
- $s \in \{0,1,2,\dots\}$: **seam count** (hairline or horizon only; props/ornaments don't count)

Ghost Density (overall feasible set)

Base envelopes

- $r_v \in [0.62, 0.90]$
- $\rho_r \in [0.18, 0.40]$

Anchor clauses

- $(e \vee h) \Rightarrow (\Delta x \in [0.12, 0.20] \wedge s = 1)$
- $(\neg e \wedge \neg h) \Rightarrow (r_v \geq 0.74 \wedge s \leq 1_{\text{faint}})$ — field-void compensation

Profiles (read-offs from the envelope)

- **GD–Edge–Anchored**: $e=1, h=0 \rightarrow \Delta x \in [0.15, 0.18], r_v \in [0.62, 0.68], \rho_r \in [0.28, 0.40], s=1$.
(Tightened Δx & r_v for reliability.)
- **GD–Horizon**: $e=0, h=1 \rightarrow$ same Δx band if a subject exists; r_v typically higher $[0.70, 0.85]$; $\rho_r \approx [0.18, 0.30]$; $s=1$ (horizon).
- **Field–Void** (unanchored): $e=0, h=0 \rightarrow$ ignore Δx for pass; demand more void: $r_v \geq 0.74$; $\rho_r \in [0.18, 0.32]$; $s \leq 1_{\text{faint}}$.

Common fail zones (explicit):

- **Snap-back collapse**: $\Delta x < 0.10$ (when evh), $r_v < 0.55$.
- **Over-void dissolution**: $r_v > 0.90$ (or $r_v > 0.87$ with no subject cue).
- **Candy-polish**: $\rho_r < 0.20$ with centered saliency/props.

Confidence envelopes (proposal framing)

Report bands as empirical envelopes, not hard cuts:

- **Edge–Anchored (artist basin)**: $\Delta x \approx 0.15$ with 95% CI $[0.12, 0.18]$; $r_v \approx 0.65$ CI $[0.62, 0.69]$; $\rho_r \approx 0.34$ CI $[0.28, 0.40]$; $s=1$.
- **Horizon**: Δx same if subject; r_v mode ≈ 0.78 CI $[0.70, 0.85]$; $\rho_r \approx 0.24$ CI $[0.18, 0.30]$; $s=1$.
- **Field–Void**: r_v mode ≈ 0.80 CI $[0.74, 0.90]$; $\rho_r \approx 0.25$ CI $[0.18, 0.32]$; $s \leq 1_{\text{faint}}$.

Engine-bias windows (deploy)

Document separately from the artist basin:

- **Midjourney (v?)**: $\Delta x + 0.02 - 0.03$ (right-drift), $\rho_r + 0.02 - 0.03$ (adds clutter). Deploy: $\Delta x \in [0.12, 0.20]$, $r_v \in [0.62, 0.68]$, $\rho_r \in [0.30, 0.42]$, seam tolerance ± 3 px.
- **Meta Image**: $r_v - 0.02$ (shrinks void), adds minor props. Deploy: bias r_v target to 0.67; forbid props explicitly.
- **SD/Firefly**: slight $+\rho_r$, horizon snap; use "single seam only; no ornament."
(These are proposal deltas; mark as such.)

Optional scalar score (soft energy)

Use as an optimizer, not a truth test; weights are illustrative:

$$\begin{aligned} E = & w_x \phi(\Delta x; \mu_x = 0.16, \sigma_x = 0.02) \cdot 1(evh) \\ & + w_v \phi(r_v; \mu_v = 0.65, \sigma_v = 0.05) \\ & + w_r \phi(\rho_r; \mu_r = 0.34, \sigma_r = 0.06) \\ & + \lambda_s |s - 1| + \lambda_o \text{ (props/ornament score)} \end{aligned}$$

For **Field–Void**, drop the Δx term and shift $\mu_v \rightarrow 0.80$.

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This framework was architected by Russell Parrish and recursively co-developed inside GPT, Gemini and Claude. Every critique is human-led; every recursion is model-driven. The result: a reasoning layer authored through language, not image manipulation.